

# JAVA PROGRAMMING LABORATORY

|                |                                              |
|----------------|----------------------------------------------|
| Experiment No. | 06                                           |
| Date           | 23/09/26                                     |
| Title          | Implementation of ArrayDeque Methods in Java |

## Objective / Aim

To write and execute a Java program that demonstrates commonly used methods of the ArrayDeque class, including addFirst(), addLast(), offerFirst(), offerLast(), peekFirst(), peekLast(), pollFirst(), and pollLast().

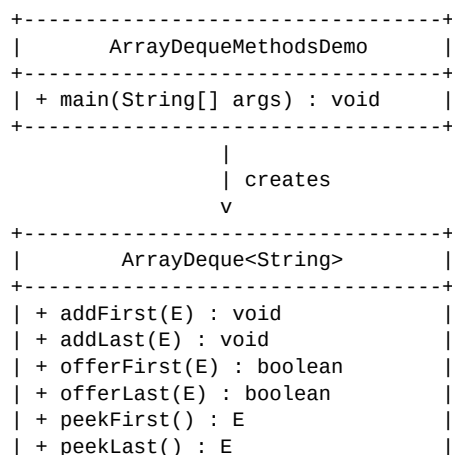
## System Requirements

| Hardware Requirements             | Software Requirements                             |
|-----------------------------------|---------------------------------------------------|
| Computer/Laptop                   | Operating System: Windows/Linux/macOS             |
| Processor: Intel Core i3 or above | JDK 17 or JDK 21                                  |
| RAM: 4 GB or above                | Java compiler (javac)                             |
| Storage: At least 1 GB free space | IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code |

## Algorithm / Logic Description

1. Import the ArrayDeque class from the java.util package.
2. Create an ArrayDeque named names.
3. Insert Rahul and Priya at the front using addFirst() and display the deque.
4. Insert Arjun and Sneha at the rear using addLast() and display the deque.
5. Insert Kiran at the front using offerFirst().
6. Insert Meena at the rear using offerLast().
7. Use peekFirst() to display the first element without removing it.
8. Use peekLast() to display the last element without removing it.
9. Use pollFirst() to remove and display the first element.
10. Use pollLast() to remove and display the last element.
11. Display the deque after each removal operation.

## UML Class Diagram



```
| + pollFirst() : E |
| + pollLast() : E |
+-----+
```

## Source Code

```
import java.util.ArrayDeque;

class ArrayDequeMethodsDemo {

    public static void main(String[] args) {

        // Creating an ArrayDeque
        ArrayDeque<String> names = new ArrayDeque<>();

        // addFirst(E e)
        names.addFirst("Rahul");
        names.addFirst("Priya");

        System.out.println("After addFirst(): " + names);

        // addLast(E e)
        names.addLast("Arjun");
        names.addLast("Sneha");

        System.out.println("After addLast(): " + names);

        // offerFirst(E e)
        names.offerFirst("Kiran");
        System.out.println("After offerFirst(): " + names);

        // offerLast(E e)
        names.offerLast("Meena");
        System.out.println("After offerLast(): " + names);

        // peekFirst()
        System.out.println("First element using peekFirst(): "
            + names.peekFirst());

        // peekLast()
        System.out.println("Last element using peekLast(): "
            + names.peekLast());

        // pollFirst()
        System.out.println("Removed first element using pollFirst(): "
            + names.pollFirst());

        System.out.println("After pollFirst(): " + names);

        // pollLast()
        System.out.println("Removed last element using pollLast(): "
            + names.pollLast());

        System.out.println("After pollLast(): " + names);
    }
}
```

## Completion and Execution Commands

```
Save as:
ArrayDequeMethodsDemo.java

Compile:
javac ArrayDequeMethodsDemo.java

Execute:
java ArrayDequeMethodsDemo
```

## Output

```
After addFirst(): [Priya, Rahul]
After addLast(): [Priya, Rahul, Arjun, Sneha]
After offerFirst(): [Kiran, Priya, Rahul, Arjun, Sneha]
After offerLast(): [Kiran, Priya, Rahul, Arjun, Sneha, Meena]
First element using peekFirst(): Kiran
```

```
Last element using peekLast(): Meena
Removed first element using pollFirst(): Kiran
After pollFirst(): [Priya, Rahul, Arjun, Sneha, Meena]
Removed last element using pollLast(): Meena
After pollLast(): [Priya, Rahul, Arjun, Sneha]
```

Note: ArrayDeque supports insertion and removal efficiently at both ends. It does not permit null elements.